

MIDTERM LEARNING GUIDE

- Listed below are the Learning Objectives for the Midterm Exam, which covers Units 1-3. In the table below write your own exam question that covers the given objective. Use this to structure your own learning guide and better focus your study time. Also determine your confidence with each objective with the numerical key below.

KEY: 1 = Not Confident 2 = Somewhat Confident
3 = Absolutely Confident

Unit Learning Objective	Create Your Own Questions	Confidence	Prior Exam Performance
Unit 1			
Use multiple representations to identify functions and justify what defines a function.			
Find function values and describe their meaning from a given scenario.			
Evaluate and solve for function values given a graph.			
Determine increasing and decreasing intervals of various functions.			
Identify domain and range of various functions using interval, set, and/or inequality notation.			
Determine the composite function from a given pair of functions and their associated domain.			
Evaluate composite functions at a constant, taking the domain into account.			
Find an inverse of various functions algebraically, restricting the domain when necessary.			

Unit Learning Objective	Create Your Own Questions	Confidence	Prior Exam Performance
Use properties of functions to explain the relationship between a function and its inverse.			
Represent rate of change graphically to/from a scenario (distance/time, bottles, etc.).			
Identify and use function patterns in given sets of data and connect to general functions.			
Unit 2			
Determine the effects of various transformations on a given parent function or a general function.			
Find attributes of quadratic functions (complex/real roots, y-intercept, vertex, etc.) from an equation.			
Algebraically manipulate different forms of conic equations to identify all key components for each conic.			
Graph conics and identify key components.			
Write the standard form of a conic (circle, ellipse, hyperbola, parabola) from key components.			
Demonstrate understanding of the locus definition of conics.			
Use matrices to model conics from given points.			
Unit 3			

Unit Learning Objective	Create Your Own Questions	Confidence	Prior Exam Performance
Simplify and evaluate exponential expressions using properties.			
Simplify and evaluate logarithmic expressions using properties.			
Expand or condense logarithmic expressions using properties.			
Compute logarithms mentally without a calculator.			
Understand and be able to use the number <i>e</i> as an exponential and logarithmic base.			
Write equations of logarithmic functions given a set of points or a graph.			
Identify situations as exponential growth or decay.			
Create and use models for exponential growth or decay from real-world scenarios (half-life, compound interest, etc.).			
Solve exponential and logarithmic equations by using their inverse relationship.			
Solve exponential and logarithmic equations by using a common base.			
Use function patterns to identify and predict values of exponential and logarithmic functions.			

2. What are your goals for the Midterm Exam? Remember that a Satisfaction Goal is the grade you would need to be satisfied with your work, and a Pride Goal is the grade necessary for you to feel pride in your work.



EXPERIENCE COLLEGE BEFORE COLLEGE

Satisfaction Goal = _____/100

Pride Goal = _____/100